الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

WEEK 6 · LESSONS 6-5 · 7-1

Weekly Quiz Preparation

Grade 11 Advanced · Mathematics

كتيّب التجهّز للاختبار الأسبوعي القصير

1 2 3 4

4	10	40	14
objectives أهداف	marks علامة	minutes دقيقة	practice questions تدريبًا

Mr. Yousef Al Yousef · **Friday 9 October 2026**

One page per objective · every solution written out step by step

Quiz map

4 objectives = 4 recipes · 4 أهداف = 4 وصفات

1 **Growth problem** النمو الأسي
y = a e^(kt): substitute, divide by a, take ln, divide by k

2 **Decay / half-life problem** الاضمحلال
k = ln 2 / half-life · fraction left = e^(−kt) · take ln

3 **Simplify a rational expression** التبسيط
Factor top and bottom → divide out → excluded values from the original

4 **Multiply or divide** الضرب والقسمة
Flip after ÷ → factor everything → divide out across the product

1 · Learn
Read the idea and the recipe
اقرأ الفكرة والوصفة

2 · Watch
Cover the solution, try it, then compare
غطِّ الحل وحاول ثم قارن

3 · Try
Do “Your turn”, check at the back
حلّ التدريبات وصحّح

1 Exponential growth

النمو الأسي

The model y = a·e^(kt): a is the starting amount, k the continuous growth rate, t the time.

In y = a e^(kt), put the numbers in, isolate the exponential, then take ln to bring t down.

في y = a e^(kt) عوّض الأرقام واعزل الأسّي ثم خذ ln لتنزل t.

Recipe الوصفة

1. **Read a and k (k = rate as a decimal)** — اقرأ a وk
2. **Substitute the target value for y** — عوّض قيمة y المطلوبة
3. **Divide by a: isolate e^(kt)** — اقسم على a
4. **Take ln of each side: t = ln(...) / k** — خذ ln ثم اقسم على k

Watch out

- Rate 4% means k = 0.04, not 4.
- Keep k unrounded until the last line.

Tool. ln e^u = u · two data points: k = (1/t) · ln(y / a)

Worked example Authored (shape of Ex 13)

A population of 500 grows continuously at a rate of 4% a year. How long will it take to reach 800?

1. a = 500 and k = 0.04

$$y = 500e^{0.04t}$$

2. put y = 800

$$800 = 500e^{0.04t}$$

3. divide both sides by 500

$$1.6 = e^{0.04t}$$

4. take ln of each side: ln e^u = u

$$\ln 1.6 = 0.04t$$

5. divide by 0.04

$$t = \frac{\ln 1.6}{0.04} \approx 11.75 \text{ years}$$

Your turn · solve in your notebook · solutions at the back

1. **Authored (shape of Ex 13)** A car costs AED 40,000 and its price rises continuously at 3% a year. When will it cost AED 50,000?
2. **Book exercise 16** A colony has 20 bacteria at 10:00 A.M. and grows continuously at 8% each hour. How many hours until it reaches 50 bacteria?
3. **Authored (shape of Ex 1)** A town had 12 million people in 2010 and 13.2 million in 2015. Write $y = ae^{kt}$ for the population t years after 2010.

2 Exponential decay
الاضمحلال الأسي

y = a·e^(−kt): the amount shrinks. Half-life h gives k = ln 2 / h.

Half-life gives k = ln 2 ÷ h; the fraction left is the value of e^(−kt).

عمر النصف يعطي k = ln 2 ÷ h والنسبة المتبقية = e^(−kt).

Recipe الوصفة

1. **Find k: half-life h ⇒ k = ln 2 / h**
   k من عمر النصف
2. **Write the fraction left (20% left = 0.20)**
   اكتب النسبة المتبقية
3. **Solve the fraction left = e^(−kt)**
   e^(−kt) = نسبة
4. **Take ln: t = ln(fraction) / (−k)**
   خذ ln

Watch out

- "80% lost" means 20% is left.
- The minus sign belongs in the formula; k itself is positive.

Tool. after one half-life half is left · Carbon-14: k = 0.00012

Worked example Authored (shape of Ex 3, 6)

The half-life of strontium-90 is 29 years. Find k, then how long until only 20% of a sample is left.

1. half-life 29 years ⇒ k = ln 2 / 29

$$k = \frac{\ln 2}{29} \approx 0.0239$$

2. 20% left: 0.2 = e^(−kt)

$$0.2 = e^{-0.0239t}$$

3. take ln of each side

$$\ln 0.2 = -0.0239t$$

4. divide by −k

$$t = \frac{\ln 0.2}{-0.0239} \approx 67.3 \text{ years}$$

Your turn · solve in your notebook · solutions at the back

1. **Book exercise 10** Carbon-14 has $k = 0.00012$. A fossil has lost 80% of its Carbon-14. How old is it?
2. **Authored (shape of Ex 3)** An 80 g sample has a half-life of 6 days. How much is left after 15 days?
3. **Authored (shape of Ex 4)** Carbon-14 has $k = 0.00012$. How old is a skeleton that has 60% of its Carbon-14 remaining?

3 Simplifying rational expressions
تبسيط العبارات النسبية

Lesson 7-1 · Part 1 · pp. 311–316
MAT.2.02.20.001

Factor everything, cancel what is common, and say which values the ORIGINAL expression forbids.

Factor the top and the bottom completely, then divide out the common factors.
حلّل البسط والمقام تحليلًا كاملًا ثم اختصر العوامل المشتركة.

Recipe الوصفة

1. **Factor the numerator completely** — حلّل البسط
2. **Factor the denominator completely** — حلّل المقام
3. **Divide out the common factors** — اختصر العوامل المشتركة
4. **Excluded values: zeros of the original denominator** — القيم المستبعدة من المقام الأصلي

Watch out

- Cancel factors, never terms: (x + 3)/(x + 5) does not become 3/5.
- Excluded values come from the ORIGINAL denominator.

Tool. a − b = −(b − a) · a² − b² = (a − b)(a + b)

Worked example Authored (shape of Ex 1)

Simplify $\dfrac{x^2+2x-15}{x^2-9}$, and state when the original expression is undefined.

1. factor the numerator

$$x^2+2x-15=(x+5)(x-3)$$

2. factor the denominator (difference of squares)

$$x^2-9=(x-3)(x+3)$$

3. divide out x − 3

$$=\frac{(x+5)(x-3)}{(x-3)(x+3)}=\frac{x+5}{x+3}$$

4. the original is undefined where its denominator is 0

$$(x-3)(x+3)=0 \Rightarrow x=3 \text{ or } x=-3$$

Your turn · solve in your notebook · solutions at the back

1. Book exercise 1 Simplify $\dfrac{x(x-3)(x+6)}{x^2+x-12}$, and state when the original expression is undefined.
2. Book exercise 2 Simplify $\dfrac{y^2(y^2+3y+2)}{2y(y-4)(y+2)}$, and state when the original expression is undefined.
3. Book exercise 7 Simplify $\dfrac{x^2-5x-14}{28+3x-x^2}$.
4. Book exercise 10 Simplify $\dfrac{16-c^2}{c^2+c-20}$.

4 Multiplying and dividing rational expressions

ضرب وقسمة العبارات النسبية

Lesson 7-1 · Part 2 · pp. 311–316
MAT.2.02.20.005

Dividing is multiplying by the reciprocal; then factor and cancel across the product.

Multiply the numerators and the denominators — but factor first and cancel across the whole product.

اضرب البسوط والمقامات، لكن حلّل أولًا واختصر عبر الضرب كله.

Recipe الوصفة

1. **Division: flip the second fraction, multiply**
   القسمة: اقلب الكسر الثاني واضرب
2. **Factor every numerator and denominator**
   حلّل كل بسط ومقام
3. **Divide out common factors (top with bottom, any fraction)**
   اختصر العوامل المشتركة
4. **Write what is left**
   اكتب الباقي

Watch out

- Flip only the fraction after the ÷ sign.
- Cancel factors, not terms; a −1 factor changes the sign.

Tool. a/b ÷ c/d = a/b · d/c · monomials: subtract the exponents

Worked example Authored (shape of Ex 24)

Simplify $\dfrac{x^2-x-6}{x^2+3x-10}\cdot\dfrac{x^2+x-20}{x^2-9}$.

1. factor all four polynomials

$$\frac{(x-3)(x+2)}{(x+5)(x-2)}\cdot\frac{(x+5)(x-4)}{(x-3)(x+3)}$$

2. common factors: x − 3 (top-left, bottom-right) and x + 5

divide out $x-3$ and $x+5$

3. what is left

$$=\frac{(x+2)(x-4)}{(x-2)(x+3)}$$

Your turn · solve in your notebook · solutions at the back

1. Book exercise 25 Simplify $\dfrac{c^2-6c-16}{c^2-d^2}\div\dfrac{c^2-8c}{c+d}$.
2. Book exercise 28 Simplify $\dfrac{9-x^2}{x^2-4x-21}\cdot\left(\dfrac{2x^2+7x+3}{2x^2-15x+7}\right)^{-1}$.
3. Authored (shape of Ex 11–16) Simplify $\dfrac{14a^3b^2}{9c}\div\dfrac{7ab}{3c^2}$.
4. Book exercise 36 A jar holds B blue and R red jelly beans. A bag of 100 red and 100 blue is added. What is the ratio of red to blue beans in the jar?

Objective 1 · Exponential growth النمو الأسي

1.1 Authored (shape of Ex 13)

A car costs AED 40,000 and its price rises continuously at 3% a year. When will it cost AED 50,000?

1. $a = 40\,000$ and $k = 0.03$

$$y = 40000e^{0.03t}$$

2. put $y = 50\,000$ and divide by 40 000

$$50000 = 40000e^{0.03t} \Rightarrow 1.25 = e^{0.03t}$$

3. take ln, divide by 0.03

$$t = \frac{\ln 1.25}{0.03} \approx 7.44 \text{ years}$$

1.2 Book exercise 16

A colony has 20 bacteria at 10:00 A.M. and grows continuously at 8% each hour. How many hours until it reaches 50 bacteria?

1. $a = 20$ and $k = 0.08$

$$y = 20e^{0.08t}$$

2. put $y = 50$ and divide by 20

$$50 = 20e^{0.08t} \Rightarrow 2.5 = e^{0.08t}$$

3. about 11 h 27 min after 10:00 A.M.

$$t = \frac{\ln 2.5}{0.08} \approx 11.45 \text{ hours}$$

1.3 Authored (shape of Ex 1)

A town had 12 million people in 2010 and 13.2 million in 2015. Write $y = ae^{kt}$ for the population t years after 2010.

1. $a = 12$ and $t = 5$ for $y = 13.2$

$$13.2 = 12e^{5k}$$

2. divide by 12

$$e^{5k} = 1.1$$

3. take ln, divide by 5: write the model

$$k = \frac{\ln 1.1}{5} \approx 0.0191, \qquad y = 12e^{0.0191t}$$

2.1 Book exercise 10

Carbon-14 has $k = 0.00012$. A fossil has lost 80% of its Carbon-14. How old is it?

1. lost 80% ⇒ 20% left

$$0.2 = e^{-0.00012t}$$

2. take ln of each side

$$\ln 0.2 = -0.00012t$$

3. divide by −0.00012

$$t = \frac{\ln 0.2}{-0.00012} \approx 13\,412 \text{ years}$$

2.2 Authored (shape of Ex 3)

An 80 g sample has a half-life of 6 days. How much is left after 15 days?

1. half-life 6 days ⇒ k = ln 2 / 6

$$k = \frac{\ln 2}{6} \approx 0.1155$$

2. put t = 15

$$y = 80e^{-0.1155(15)}$$

3. about 14.1 g (2.5 half-lives)

$$y \approx 14.1 \text{ g}$$

2.3 Authored (shape of Ex 4)

Carbon-14 has $k = 0.00012$. How old is a skeleton that has 60% of its Carbon-14 remaining?

1. 60% left = 0.6

$$0.6 = e^{-0.00012t}$$

2. take ln of each side

$$\ln 0.6 = -0.00012t$$

3. divide by −0.00012

$$t = \frac{\ln 0.6}{-0.00012} \approx 4257 \text{ years}$$

3.1 Book exercise 1

Simplify $\frac{x(x-3)(x+6)}{x^2+x-12}$, and state when the original expression is undefined.

1. factor the denominator

$$x^2+x-12=(x+4)(x-3)$$

2. divide out x − 3

$$\frac{x(x-3)(x+6)}{(x+4)(x-3)}=\frac{x(x+6)}{x+4}$$

3. zeros of (x + 4)(x − 3)

undefined at $x=-4,\ x=3$

3.2 Book exercise 2

Simplify $\frac{y^2(y^2+3y+2)}{2y(y-4)(y+2)}$, and state when the original expression is undefined.

1. factor the trinomial

$$y^2+3y+2=(y+1)(y+2)$$

2. divide out y and y + 2

$$\frac{y\cdot y\,(y+1)(y+2)}{2y(y-4)(y+2)}=\frac{y(y+1)}{2(y-4)}$$

3. zeros of 2y(y − 4)(y + 2)

undefined at $y=0,\ y=4,\ y=-2$

3.3 Book exercise 7

Simplify $\frac{x^2-5x-14}{28+3x-x^2}$.

1. factor; the denominator has a leading −1

$$x^2-5x-14=(x-7)(x+2),\qquad 28+3x-x^2=-(x-7)(x+4)$$

2. divide out x − 7: keep the minus sign

$$\frac{(x-7)(x+2)}{-(x-7)(x+4)}=-\frac{x+2}{x+4}$$

3.4 Book exercise 10

Simplify $\frac{16-c^2}{c^2+c-20}$.

1. factor: $16-c^2=-(c-4)(c+4)$

$$16-c^2=-(c-4)(c+4),\qquad c^2+c-20=(c+5)(c-4)$$

2. divide out c − 4: keep the minus sign

$$\frac{-(c-4)(c+4)}{(c+5)(c-4)}=-\frac{c+4}{c+5}$$

Objective 4 · Multiplying and dividing rational expressions — ضرب وقسمة العبارات النسبية

4.1 Book exercise 25

Simplify $\dfrac{c^2-6c-16}{c^2-d^2} \div \dfrac{c^2-8c}{c+d}$.

1. flip the second fraction; factor $c^2-6c-16$, c^2-d^2, c^2-8c

$$\frac{(c-8)(c+2)}{(c-d)(c+d)} \cdot \frac{c+d}{c(c-8)}$$

2. divide out $c-8$ and $c+d$

$$= \frac{c+2}{c(c-d)}$$

4.2 Book exercise 28

Simplify $\dfrac{9-x^2}{x^2-4x-21} \cdot \left(\dfrac{2x^2+7x+3}{2x^2-15x+7}\right)^{-1}$.

1. the power -1 flips the second fraction; factor everything

$$\frac{-(x-3)(x+3)}{(x-7)(x+3)} \cdot \frac{(2x-1)(x-7)}{(2x+1)(x+3)}$$

2. divide out $x+3$ (once) and $x-7$

$$= -\frac{(x-3)(2x-1)}{(2x+1)(x+3)}$$

4.3 Authored (shape of Ex 11–16)

Simplify $\dfrac{14a^3b^2}{9c} \div \dfrac{7ab}{3c^2}$.

1. flip the second fraction

$$\frac{14a^3b^2}{9c} \cdot \frac{3c^2}{7ab}$$

2. subtract exponents: $a^3/a = a^2$, $b^2/b = b$, $c^2/c = c$; $42/63 = 2/3$

$$= \frac{42a^3b^2c^2}{63abc} = \frac{2a^2bc}{3}$$

4.4 Book exercise 36

A jar holds B blue and R red jelly beans. A bag of 100 red and 100 blue is added. What is the ratio of red to blue beans in the jar?

1. red beans: $R+100$ · blue beans: $B+100$ · the ratio is red : blue

$$\frac{R+100}{B+100}$$

Ready?

Tick each line · ضع علامة

- [ ] **I can solve a growth problem with y = a e^(kt)**
  أستطيع حل مسائل النمو الأسي
- [ ] **I can find k from a half-life and solve a decay problem**
  أستطيع حل مسائل الاضمحلال
- [ ] **I can simplify a rational expression and state when it is undefined**
  أستطيع تبسيط العبارة النسبية
- [ ] **I can multiply and divide rational expressions**
  أستطيع ضرب وقسمة العبارات النسبية

On the day

Bring a scientific calculator. Write the first line at once: the first step already earns a mark.

Every question

Factor completely before you cancel. Keep k unrounded until the last line. State the excluded values when asked.

One mark for every written step

لكل خطوة تكتبها علامة